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Machine learning approaches to urban heat Island detection and sustainability analysis

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Abstract

Urban Heat Islands (UHIs) represent a critical barrier to sustainable urbanism and ecological stability in rapidly expanding metropolitan regions. A significant gap exists in operationalising these data for governance and climate-resilient planning, even though remote sensing of heat is a well-established approach. This study proposes a comprehensive conceptual framework for leveraging Machine Learning (ML) to bridge the divide between high-resolution thermal detection and sustainability-focused urban policy. The paper outlines a methodological roadmap for integrating large-scale geospatial datasets, including Land Surface Temperature (LST) and urban morphology into predictive ML architectures such as Random Forest and Convolutional Neural Networks (CNNs), with the synthesis of key interdisciplinary literature. The research moves beyond descriptive analysis to explore the governance-oriented insights of AI, specifically in prioritising green infrastructure expansion and ecological corridor conservation for urban biodiversity. A critical examination of Explainable AI (XAI) is provided to address the trade-offs between model accuracy and the transparency required for public accountability in urban design. This work presents AI-driven UHI mapping not merely as a technical exercise, but as a strategic catalyst for achieving Sustainable Development Goal 11 (SDG 11) to make cities and human settlements inclusive, safe, resilient, and sustainableThe key findings supports a paradigm shift toward interdisciplinary, AI-informed urban management to preserve the ecological resilience against the dual pressures of climate change and accelerated urbanisation.

Keywords Artificial intelligence, Energy, Green infrastructure, Planning, Urban climate

1 Introduction

Urbanisation is an essential component of development in the of the modern era, and the rapid urban growth has led to the emergence of urban heat islands (UHIs), where cities experience significantly higher temperatures compared to their surrounding rural areas [44, 46]. UHIs result from the absorption and retention of solar radiation by built surfaces, along with human activities such as industrialisation, transportation, and energy consumption [31, 52, 79]. These higher temperatures aggravate heat-related health risks, increase energy demand for cooling, and disrupt ecological systems, posing significant challenges to sustainable development and ecological conservation. Figure 1

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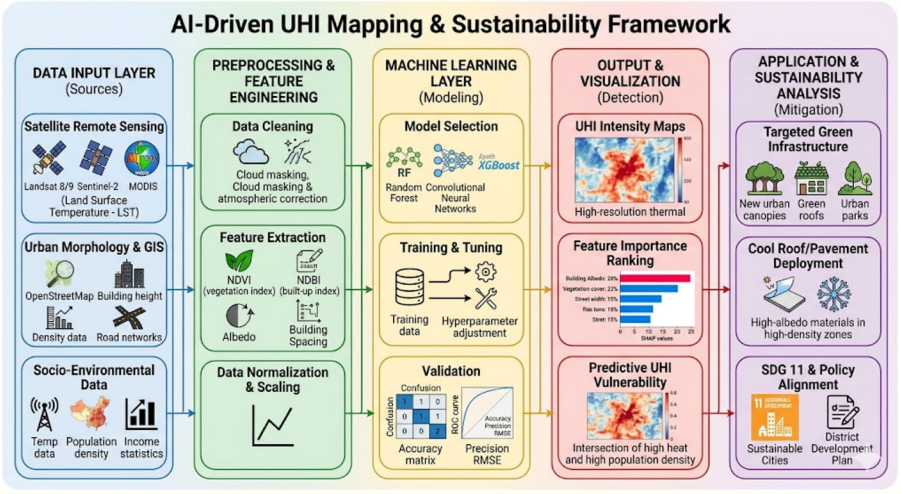


Fig. 1 Conceptual framework for AI-Driven UHI Analysis & Sustainable Mitigation Workflow. *(Source: Compiled by the authors with the assistance of AI-based tools for conceptual visualization and design enhancement)*

shows a detailed conceptual/theoretical framework for an AI-driven Urban Heat Island (UHI) mapping and sustainability analysis workflow. The figure has following outlined components:

1. **Input Layer (The raw data sources):** This is required as a foundational building block required for the detection algorithm.

   - **Geospatial & Remote Sensing:** Landsat 8/9, MODIS, and Sentinel-2 data (for multi-temporal Land Surface Temperature/LST analysis).
   - **Urban Morphology:** NDBI (Normalised Difference Built-up Index), Building Density, Road Networks (derived from OpenStreetMap or other spatial datasets).
   - **Socio-Environmental Data:** Population Density and Population Income statistics (to identify vulnerable demographics).
   - **Weather Data:** Temperature logs and NDVI (Normalised Difference Vegetation Index) data.

2. **Preprocessing & Feature Engineering (The technical data prep):** It shows technical expertise in data quality control.

   - **Cloud Masking & Atmospheric Correction:** Demonstrating that your data is clean and consistent across temporal scales (using platforms like Google Earth Engine).
   - **Feature Selection:** Highlighting the indices used (NDVI, NDBI) and any Principal Component Analysis (PCA) performed to identify the key drivers of localised heat.

3. **Processing Layer (The core Machine Learning engine):** It deploys the AI models described in the methodology. A flowchart of a machine learning data pipeline with preprocessing, model training, and performance evaluation.

   - Model Selection: Specifically names Random Forest (RF), Convolutional Neural Networks (CNN), and XGBoost.

- Training & Tuning: Shows the data split and hyperparameter adjustment process.
- Accuracy Assessment: Includes placeholders for R^2 and RMSE (Root Mean Square Error) validation metrics.

4. **Output Layer (The actionable detection results)**

- **UHI Intensity Maps:** Identifying the precise location of thermal hotspots.
- **Predictive UHI Vulnerability:** Forecasting future hotspots based on urban growth projections.
- **SHAP (SHapley Additive exPlanations) Feature Ranking:** A novel technical addition that quantifies which urban features contribute most significantly to localised heat (e.g., *Albedo: 25%, NDVI: 20%*).

5. **Application Layer (The "Sustainability Analysis" component):** Connecting detection to planning solutions (SDG 11). Urban Heat Island mitigation strategies involving cool roofs and blue-green infrastructure.

- **Green Infrastructure Prioritisation:** Targeting hotspots for urban canopies, green roofs, and blue spaces (water bodies).
- **Cool Roof/Pavement Deployment:** Recommending high-albedo materials for high-density, low-income urban zones.
- **Policy & Policy Alignment:** Integrating UHI maps into District Development Plans and SDG 11 (Sustainable Cities and Communities) goals.

The UHI impact requires unconventional solutions to address the problems with advanced technologies and interdisciplinary approaches. Artificial intelligence (AI) in recent years has emerged as a powerful tool for mapping and analysing UHI [60, 71]. It offers insights into spatial distribution, intensity, and underlying drivers. AI algorithms process vast amounts of geospatial data, including satellite imagery, environmental sensors, and urban infrastructure data, to create detailed maps of UHIs at high spatial and temporal resolutions [78]. Fig. 1 presents the conceptual framework for the AI-driven Urban Heat Island (UHI) analysis and sustainable mitigation workflow. The framework illustrates the overall methodology adopted in the study, beginning with data collection from remote sensing, climatic, land-use, and geospatial sources. It further demonstrates how artificial intelligence and machine learning techniques are integrated to process, analyze, and model UHI patterns across urban environments. The workflow also highlights the identification of heat-vulnerable zones, assessment of environmental impacts, and development of sustainable mitigation strategies, such as green infrastructure planning, urban vegetation enhancement, and climate-responsive urban design, to support resilient and sustainable urban development.The potential of AI-motivated UHI mapping for promoting sustainable development and ecological conservation is used in urban environments. In precise identification of UHI hotspots for understanding their spatial dynamics, policymakers and urban planners prioritise interventions to mitigate heat-related health risks and enhance energy efficiency to optimise green infrastructure planning [6, 56, 76]. AI-enabled UHI mapping communicates about ecological conservation efforts for the identification of areas where high temperatures negatively impact urban biodiversity, wildlife habitats, and ecological corridors.

The UHI mapping with the AI involves interdisciplinary collaboration among scientists, policymakers, urban planners, and communities. Cities foster sustainable development to safeguard ecological resilience in the aspect of climate change and rapid urbanisation with the integration of AI-driven solutions to urban planning and management processes [29, 77]. This provides insights into the applications of AI for UHI mapping and its implications for sustainable urban development and ecological conservation.

AI contribute in addressing UHI and promoting sustainable development and ecological conservation [7, 120]. Mapping urban heat islands (UHIs) with artificial intelligence (AI) embraces the substantial potential for sustainable development and ecological conservation efforts, including:

a) *High-resolution Mapping:* AI algorithms can analyse large volumes of satellite imagery and other geospatial data to create high-resolution maps of urban heat islands. These maps can identify hotspots and quantify temperature variations across urban areas, providing valuable insights for urban planning and land use management.
b) *Predictive Modelling:* AI techniques, such as machine learning algorithms, develop predictive models to forecast UHI intensity and spatial distribution under different scenarios, such as changes in land use, climate conditions, or urban development patterns. These models help policymakers and urban planners make informed decisions to mitigate UHIs and adapt to climate change.
c) *Identifying Vulnerable Populations:* AI-powered UHI mapping identifies areas with high concentrations of vulnerable populations like the elderly, children, or low-income communities. These are more susceptible to heat-related health risks. This information guides targeted interventions and resource allocation to protect public health and enhance community resilience.
d) *Green Infrastructure Planning:* AI contributes to optimising the design and placement of green infrastructure, such as parks, green roofs, and urban forests, for effective UHI mitigation. The analysis of environmental data and urban heat patterns with AI algorithms recommends strategic locations for green spaces to maximise the cooling benefits and enhance ecological conservation.
e) *Energy Efficiency:* AI-driven UHI mapping updates energy-efficient building design and urban infrastructure planning. The identification of areas with high heat intensity supports decision-makers in prioritising investments in energy-efficient technologies, such as cool roofs, reflective pavements, and natural ventilation systems for reducing energy consumption and combating urban heat.
f) *Ecological Conservation:* Understanding the spatial distribution of UHIs is crucial for conserving urban biodiversity and ecosystems. AI analyses the impact of UHI on wildlife habitats, species distribution, and ecological corridors within cities. The integration of ecological data with UHI mapping helps ecologists advance strategies for conserving urban biodiversity and enhancing ecosystem resilience in the face of climate change.

AI for mapping urban heat islands in fast-urbanising areas helps in sustainable urban growth, better public health, and more environmental conservation [5, 94]. It manages the complicated issues related to UHI and executes AI-driven solutions for interdisciplinary collaboration among communities, scientists, politicians, and urban planners. Fig. 2 presents the conceptual framework for the proposed AI-driven Urban Heat Island

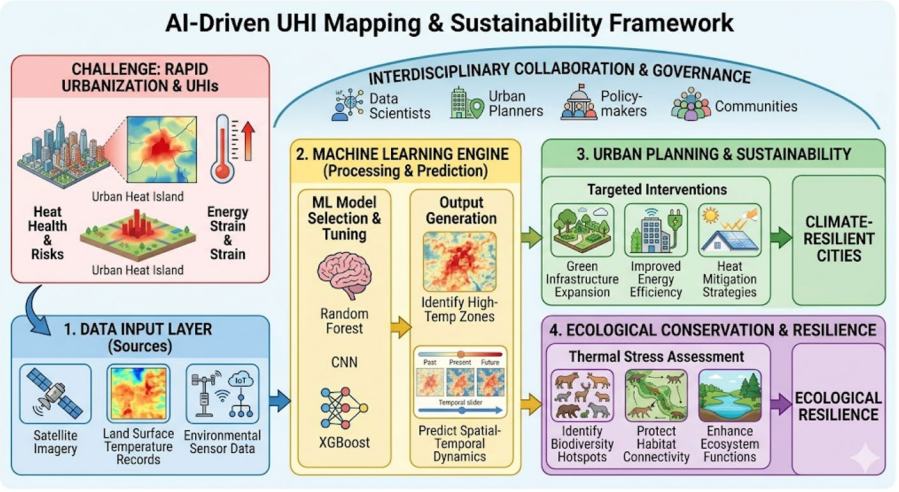


Fig. 2 Conceptual framework for AI-Driven UHI Mapping & Sustainability. *(Source: Compiled by the authors with the assistance of AI-based tools for conceptual visualization and design enhancement)*

(UHI) mapping and sustainability framework. The diagram illustrates the integration of artificial intelligence techniques, spatial and environmental datasets, and analytical processes used to identify, map, and assess UHI patterns. It also highlights the relationship between data acquisition, model development, spatial analysis, and sustainability-oriented decision-making for urban planning and climate resilience.

The figure 2 illustrates a multi-stage Conceptual Framework for AI-Driven UHI Mapping and Sustainability to provide a methodological roadmap for transitioning from raw data to urban resilience. The process begins with a Data Input Layer that aggregates multi-source geospatial information, including satellite imagery (LST), urban morphology (building height/density), and socio-environmental statistics. These data undergo Preprocessing and Feature Engineering to ensure atmospheric correction and the extraction of key indices like NDVI and NDBI. The core framework is based onthe Machine Learning Engine, where ensemble models like Random Forest and CNNs are trained and validated to generate high-resolution UHI Intensity Maps and predictive vulnerability assessments. The framework bifurcates these analytical outputs into two actionable streams: *Urban Planning and Sustainability*, which focuses on targeted interventions like green infrastructure and energy efficiency for SDG 11 alignment, and *Ecological Conservation*, which prioritises habitat connectivity and biodiversity protection against thermal stress. This entire pipeline is supported by a governance layer of interdisciplinary collaboration among data scientists, planners, and policymakers.

Defining the Thermal Scope: Surface vs. Atmospheric UHI provide methodological clarity, thus it is essential to distinguish between the two primary forms of the Urban Heat Island effect:

- **Surface UHI (S-UHI):** Derived from thermal infrared sensors, S-UHI measures the Land Surface Temperature (LST) of the "urban skin" (roofs, pavements, and vegetation). It is characterized by high spatial resolution and is most intense during the day when surfaces absorb solar radiation.
- **Atmospheric UHI (A-UHI):** Measured via weather stations or mobile traverses, A-UHI reflects Air Temperature within the urban canopy layer. It typically peaks

at night due to the slow release of heat from urban materials and is influenced by anthropogenic heat and wind patterns.

This theoretical/ conceptual study adopts S-UHI as the primary detection metric for several reasons:

1. **Spatial Continuity:** Satellite-derived LST provides wall-to-wall spatial coverage, which is essential for training spatially aware ML models like Convolutional Neural Networks (CNNs).
2. **Actionable Planning:** S-UHI is a direct proxy for urban surface properties (Albedo, NDVI). This makes it highly relevant for "Cool Roof" interventions and green infrastructure planning, which are central to the sustainability analysis in this work.

1.1 Mapping urban heat Islands for sustainable development and ecological conservation

Mapping of urban heat islands (UHIs) is crucial for advancing sustainable development and ecological conservation efforts in urban areas. Thismapping involves various technologies and methodologies to identify hotspots, understand spatial patterns, and inform mitigation strategies. The key areas contributing to sustainable development and ecological conservation include:

a) *Identifying Vulnerable Areas:* It helps to identify areas within cities that are most susceptible to heat-related health risks, predominantly for vulnerable populations such as the elderly, children, and low-income communities. Pinpointing these vulnerable areas helps policymakers to prioritise interventions to protect public health and enhance community resilience.
b) *Optimising Urban Planning:* It provide valuable insights for urban planners to optimise land use and infrastructure development. The understanding of UHIs is most intense for planners to incorporate strategies such as green spaces, cool roofs, and reflective surfaces to mitigate heat buildup and thermal comfort enhancement for residents.
c) *Promoting Energy Efficiency:* It facilitates the identification of areas with high energy demand for cooling. The energy-efficient interventions in these hotspots, such as improving building insulation and promoting the use of passive cooling techniques, help to reduce energy consumption and greenhouse gas emissions, contributing to climate change mitigation efforts.
d) *Conserving Biodiversity:* UHIs disrupt ecological systems and biodiversity within urban environments. Mapping helps to identify areas of temperature extremes, which are a threat to native flora and fauna. The integration of mapping into ecological conservation strategies prioritises the preservation of green spaces, urban forests, and wildlife habitats to maintain biodiversity and ecosystem services.
e) *Enhancing Resilience to Climate Change:* Understanding the spatial distribution is essential for building resilience to climate change impacts. UHI maps advises the climate adaptation strategies by highlighting areas prone to heat stress and informing the design of heat-resilient infrastructure and urban greenways.

It is impossible to make progress in the areas of sustainable urban development and environmental protection without adequate mapping of urban heat islands. The mapping helps in the identification of hotspots, establishing the framework for urban planning decisions, the promotion of energy efficiency, the preservation of biodiversity, and

the enhancement of climate resilience to create cities healthier and more habitable for future generations to come [111, 115].

1.2 Key challenges in mapping urban heat Islands for sustainable development and ecological conservation

Mapping urban heat islands (UHIs) for sustainable development and ecological conservation provides several key challenges that need to be addressed for effective mitigation of the adverse impacts of heat islands to promote urban resilience [91]. Some of these challenges include:

a) *Data Availability and Quality:* Access to high-resolution, up-to-date, and consistent data on land cover, land use, surface temperature, and other relevant variables is essential for accurate UHI mapping. However, data availability and quality can vary across regions, hindering comprehensive and standardised assessments of UHIs.
b) *Spatial and Temporal Resolution:* Mapping UHI requires data with adequate spatial and temporal resolution to capture fine-scale variations in temperature and land cover within urban areas. Obtaining such high-resolution data can be challenging, particularly in densely populated cities or areas with limited access to remote sensing technologies.
c) *Data Integration and Interdisciplinary Collaboration:* UHI mapping involves integrating data from multiple sources, including remote sensing, meteorological stations, and ground-based measurements. Achieving seamless data integration often requires collaboration across disciplines, such as climatology, urban planning, ecology, and public health, which can pose coordination challenges.
d) *Urban Morphology and Heterogeneity:* The complex morphology and heterogeneity of urban environments pose challenges for accurately characterising UHIs. Variations in building density, land cover types, surface materials, and anthropogenic heat sources can influence local temperature patterns, requiring sophisticated modelling approaches to account for spatial heterogeneity.
e) *Modelling Uncertainty and Validation:* UHI mapping techniques comprise various modelling assumptions and uncertainties that can affect the accuracy and reliability of results. Robust validation methods are essential for assessing the performance of UHI models and understanding their limitations, yet validating UHI mapping results can be challenging due to the lack of ground truth data at sufficient spatial and temporal scales.
f) *Equity and Social Justice:* UHI mapping efforts must consider the social and equity implications of heat exposure, as vulnerable populations are often disproportionately affected by UHI effects. Ensuring that UHI mitigation strategies benefit all communities requires addressing underlying social inequalities and promoting inclusive urban planning processes.
g) *Long-Term Monitoring and Adaptation:* UHIs are dynamic phenomena influenced by seasonal variations, land use changes, and climate trends. Long-term monitoring and adaptive management strategies are needed to track changes in UHI intensity over time, assess the effectiveness of mitigation measures, and adapt to evolving urban and climatic conditions.

Multidisciplinary teams, new technologies, data-sharing programs, and community involvement are required to tackle these problems. Stakeholders can enhance the accuracy of UHI mapping to support more effective mitigation strategies and advance sustainable urban planning and development worldwide.

1.3 Strategies for mapping urban heat islands for sustainable development and ecological conservation

Mapping urban heat islands (UHIs) for sustainable development and ecological conservation requires a comprehensive approach to integrate various strategies and methodologies. Several key strategies for effective mapping support sustainable development and ecological conservation include:

a) *Remote Sensing and Satellite Imagery:* Utilise remote sensing data and satellite imagery to capture spatial variations in surface temperature across urban areas. Thermal infrared sensors onboard satellites deliver valuable information for UHI mapping at different spatial and temporal scales.
b) *High-Resolution Spatial Data:* Access high-resolution spatial data on land cover, land use, impervious surfaces, vegetation cover, and building characteristics. The integration of data layers into UHI mapping analyses captures fine-scale variations in surface temperature and land cover composition.
c) *Advanced GIS Techniques:* Geographic Information Systems (GIS) techniques are utilised to analyse and visualise spatial data layers, identify the UHI hotspots, and quantify temperature variations. GIS allows spatial interpolation, overlay analysis, and spatial modelling to characterise UHI patterns and dynamics.
d) *Machine Learning and Artificial Intelligence:* Harness machine learning and artificial intelligence algorithms to advance predictive models for UHI mapping. AI models are trained with historical temperature data, land cover information, and meteorological variables to forecast UHI intensity and spatial distribution under different scenarios.
e) *Ground-Based Measurements:* Remote sensing data with ground-based measurements of temperature with weather stations, thermal sensors, and handheld infrared cameras. Field surveys are conducted to validate UHI mapping results to calibrate remote sensing models for improved accuracy.
f) *Community Engagement and Citizen Science:* Engage local communities and citizen scientists in UHI mapping efforts by crowdsourcing temperature data, conducting participatory mapping exercises, and raising awareness about the UHI impacts on public health and the environment.
g) *Integrated Planning and Management:* Integration of UHI mapping with urban planning and management processes to inform decision-making and policy development. Collaboration of municipal authorities, urban planners, and environmental agencies to prioritise UHI mitigation strategies, such as increasing vegetation cover, promoting cool roofs, and enhancing green spaces.
h) *Ecological Conservation Considerations:* A Combination of ecological conservation considerations with UHI mapping efforts by assessing the impacts of UHIs on urban biodiversity, wildlife habitats, and ecosystem services. Identification of areas with UHI mitigation measures to enhance ecological resilience and promote the conservation of urban green spaces, riparian corridors, and natural habitats.

i) *Long-Term Monitoring and Evaluation:* Establish long-term monitoring programs to track changes in UHI intensity, evaluate the effectiveness of mitigation measures, and adapt management strategies to evolving urban and climatic conditions. Implementation of continuous monitoring systems with remote sensing technologies and weather stations to assess UHI trends over time.

Stakeholders can map urban heat islands accurately using a collaborative, cross-disciplinary approach.This will support sustainable development and ecological conservation goals, boost resilience to climate change, and increase the quality of life in urban areas.

1.4 Mapping urban heat Islands with artificial intelligence

Urban heat islands (UHIs) present a pressing challenge in urban environments, impacting public health, energy consumption, and ecological balance. UHI Mapping with artificial intelligence (AI) offers a creative approach to understanding and mitigating these effects . AI approaches such as machine learning algorithms, enable the analysis of huge geospatial data with satellite imagery, land cover information, and meteorological data to identify UHI patterns with the prediction of their spatial distribution with precision. UHI mapping becomes more efficient, scalable, and cost-effective compared to traditional methods with the AI approach [89]. AI systems extract complex patterns and relationships from heterogeneous datasets to provide valuable insights into the drivers and dynamics of UHI formation [66, 122]. UHI mapping with AI enables real-time monitoring and adaptive management strategies to respond to changing urban and climatic conditions. The application of AI in UHI mapping extends beyond scientific research to inform evidence-based decision-making and policy development. Urban planners, policymakers, and environmental agencies utilise AI-generated UHI maps to prioritise interventions, optimise land use planning, and design resilient urban infrastructure [70, 72]. AI-driven UHI mapping contributes to broader sustainability goals by promoting energy efficiency, enhancing public health outcomes, and conserving urban biodiversity. Collaborative efforts among researchers, practitioners, and community stakeholders are essential to address these challenges and ensure that AI-driven UHI mapping contributes to creating healthier, more resilient, and sustainable cities for all [15, 48, 88]. Although AI offers significant potential for advancing UHI mapping includes several challenges, including data availability, model interpretability, and equity considerations.

Urban heat island mapping with artificial intelligence represents a powerful tool for understanding, managing, and mitigating the impacts in urban environments [50, 56, 62]. The capabilities of AI propose new insights and solutions to address complex challenges and promote sustainable urban development with ecological conservation [39, 59]. The process of mapping urban heat islands (UHIs) with the utilisation of artificial intelligence (AI) requires the application of sophisticated algorithms and machine learning methodologies to examine extensive datasets and derive valuable insights into temperature fluctuations inside urban regions. The typical normal procedure includes:

a) *Data Collection:* The first step in UHI mapping with AI involves gathering relevant data sources, such as satellite imagery, aerial photographs, land use/land cover data, topographic information, and meteorological data. These datasets provide valuable information about surface temperature, land cover types, building density, and other factors that influence UHI formation.

b) *Preprocessing:* This ensures that the data are compatible with the machine learning models and free from errors or inconsistencies that could affect the accuracy of the analysis. Preprocessing steps are necessary to clean, normalise, and format the datasets before feeding the data into AI algorithms.
c) *Feature Extraction:* AI algorithms extract features or variables from the input data that are relevant for predicting UHI intensity and spatial distribution. This may involve identifying key characteristics such as impervious surface coverage, vegetation density, building height, and proximity to water bodies, which influence surface temperature patterns.
d) *Model Training:* Machine learning models, such as neural networks, support vector machines, or random forests, are trained using labelled training data to learn the relationship between input features and UHI characteristics. During training, the model adjusts its parameters to minimise the difference between predicted and observed UHI patterns.
e) *Model Evaluation:* Once trained, the AI model is evaluated using validation data to assess its performance and generalisation ability. Evaluation metrics such as accuracy, precision, recall, and F1 score are used to quantify the model's predictive accuracy and identify any areas for improvement.
f) *UHI Mapping:* After successful training and validation, the trained AI model is applied to map UHIs across larger geographic areas or at different spatial and temporal resolutions. The model predicts UHI intensity and spatial distribution based on input features extracted from satellite imagery or other data sources.
g) *Post-Processing and Interpretation:* The UHI mapping results undergo post-processing steps to refine the outputs, visualise the spatial patterns, and interpret the findings. GIS tools and spatial analysis techniques are used to generate maps, identify UHI hotspots, and extract actionable insights for urban planning and management.
h) *Integration and Application:* The UHI maps are generated with AI techniques and are integrated into urban planning, climate adaptation, and environmental management efforts to inform decision-making and prioritise interventions for mitigating heat-related risks and promoting sustainable development.

The use of AI improves the efficiency, accuracy, and scalability of urban heat island mapping. This provides benefits to the stakeholders in understanding the complexities of heat islands in urban settings and developing strategies to address them.

1.5 Benefits of mapping urban heat Islands with Artificial Intelligence

Mapping urban heat islands (UHIs) with artificial intelligence (AI) provides several benefits across various domains, including urban planning, public health, energy efficiency, and ecological conservation [97, 112]. Some of the key benefits of utilising AI for UHI mapping include:

a) *Improved Accuracy and Precision:* AI algorithms analyse large volumes of geospatial data with high accuracy and precision. It enables more comprehensive and reliable mapping compared to traditional methods. This enhanced accuracy allows for better identification of UHI hotspots and spatial patterns to facilitate targeted interventions and resource allocation.

b) *Efficient Data Processing:* AI techniques, such as machine learning and deep learning, enable rapid processing and analysis of diverse datasets, including satellite imagery, meteorological data, and land cover information. This efficiency streamlines the UHI mapping process, reducing time and resource requirements while increasing scalability and coverage.
c) *Real-time Monitoring and Prediction:* AI-powered UHI mapping provides real-time monitoring and prediction of temperature variations within urban areas. By continuously analysing incoming data streams, AI algorithms can detect changes in UHI intensity and spatial distribution, allowing for timely interventions and adaptive management strategies.
d) *Identification of Hotspots and Vulnerable Areas:* AI-driven UHI mapping identifies UHI hotspots and vulnerable areas within cities, where temperatures are significantly elevated and heat-related risks are more pronounced. This information helps prioritise interventions to protect vulnerable populations, such as the elderly and low-income communities, from heat-related health hazards.
e) *Optimisation of Urban Planning and Design:* UHI maps generated with AI enlighten urban planning to optimise the layout of green spaces, buildings, and infrastructure. The identification of areas with high UHI intensity helps urban planners to incorporate cooling strategies, such as green roofs, urban forests, and cool pavements, to mitigate heat buildup and enhance thermal comfort for residents.
f) *Enhanced Energy Efficiency:* AI-driven UHI mapping supports energy efficiency initiatives for reducing energy consumption in buildings and urban infrastructure. The targeted interventions in UHI hotspots, such as improving building insulation and implementing cool roofing technologies, reduce energy demand for cooling, lower greenhouse gas emissions, and enhance sustainability.
g) *Support for Climate Adaptation and Resilience:* UHI Mapping with AI influences climate adaptation efforts by providing insights into the spatial distribution of heat stress within urban areas. This information helps cities develop resilience strategies to mitigate the impacts of extreme heat events, protect critical infrastructure, and enhance community resilience to climate change.

Cities can improve their understanding, management, and mitigation of UHI effects by utilising AI abilities ([58, 109]). This will lead to sustainable development and an improvement in citizens' quality of life. One of the several advantages of utilising AI to map heat islands in urban areas includes the fact that it improves accuracy and efficiency, leads to smarter judgements, and makes cities more resilient.

1.6 Challenges of artificial intelligence in mapping urban heat Islands

Artificial intelligence (AI) extends immense potential for urban heat islands (UHIs) mapping with unprecedented accuracy and efficiency [69, 78]. However, several challenges must be addressed to fully harness the potential of AI. These challenges span technical, methodological, and ethical dimensions, and their resolution is crucial for advancing our understanding of UHIs and implementing effective mitigation strategies. Some challenges in this domain include:

a) *Data Availability and Quality:* AI algorithms require large volumes of high-quality data for training and validation. However, obtaining comprehensive and reliable datasets,

including satellite imagery, meteorological data, and ground truth measurements of surface temperature, is quite challenging, particularly in developing regions or areas with limited access to data sources.

b) *Data Heterogeneity and Integration:* Integrating heterogeneous data sources, such as remote sensing imagery, urban infrastructure data, and socioeconomic indicators, poses challenges due to differences in spatial resolution, format, and compatibility. Matching diverse datasets and developing robust data integration pipelines are critical for accurate UHI mapping with AI.

c) *Model Interpretability and Explainability:* Deep learning algorithms, such as convolutional neural networks (CNNs), are often used for UHI mapping due to their ability to extract complex patterns from spatial data. However, these models are inherently black-box and lack interpretability, making it difficult to understand the factors driving UHI predictions and assess model reliability.

d) *Bias and Equity Considerations:* AI models trained on biased or incomplete datasets may produce inaccurate or unfair predictions, particularly concerning vulnerable populations disproportionately affected by UHIs. Ensuring equity and fairness in AI-driven UHI mapping requires careful attention to data representation, model training, and validation methodologies to mitigate biases and disparities.

e) *Generalisation to New Environments:* AI models trained on data from specific geographic regions or climatic conditions may struggle to generalise to new environments with different characteristics. Transfer learning techniques and domain adaptation methods can help improve model generalisation and robustness across diverse urban settings, but they require careful calibration and validation.

f) *Scale and Computational Resources:* AI algorithms for UHI mapping often require significant computational resources, including high-performance computing infrastructure and specialised hardware accelerators (e.g., GPUs). Scaling AI models to process large-scale datasets and high-resolution imagery can be computationally intensive and resource-prohibitive for some organisations or research groups.

g) *Validation and Model Evaluation:* Assessing the performance and reliability of AI models for UHI mapping requires robust validation methodologies and ground truth data for comparison. However, obtaining accurate ground truth measurements of surface temperature at sufficient spatial and temporal scales for validation purposes can be logistically challenging and costly.

Researchers, politicians, urban planners, and community members must work together across disciplines to solve these problems. Artificial intelligence (AI)driven UHI mapping has the potential to provide more egalitarian, scalable, and accurate solutions to urban heat problems and sustainable urban development by resolving methodological, ethical, and technical issues.

1.7 Empirical case studies of ML in UHI mapping

Recent applications demonstrate the high fidelity of specific ML architectures in varied urban contexts:

- **Random Forest (RF) in Semi-Arid Climates:** Studies in cities like Delhi, India, have utilized RF to rank the influence of land-cover variables, finding that albedo

and building density often outweigh vegetation indices in predicting night-time UHI intensity.

- **CNNs for Complex Urban Morphology:** In high-density cities such as Hong Kong, Convolutional Neural Networks (CNNs) have been successfully deployed to process 3D urban canopy data. Unlike linear models, CNNs effectively capture the "street canyon" effect, improving surface temperature prediction accuracy by up to 15% compared to traditional regression.
- **XGBoost for Temporal Forecasting:** Research in London has operationalised Gradient Boosting models to forecast thermal vulnerability, allowing planners to predict heatwave impacts 48 h in advance based on historical satellite and sensor data integration.

1.8 Novelty and theoretical contribution

While the current work has extensively reviewed the evolution of machine learning (ML) algorithms for remote sensing, and it provides a significant "implementation gap" between high-resolution thermal detection and actionable urban policy. This work moves beyond the descriptive synthesis of AI capabilities to offer three distinct novel contributions:

A. The Sustainability-Governance Interface

This work proposes a governance-oriented framework unlike previous reviews that focus primarily on algorithmic performance (e.g., RMSE and R^2 values). It uniquely maps ML outputs such as SHAP-based feature importance directly to specific Sustainable Development Goal (SDG) 11 targets. It provides a roadmap for planners to translate "black-box" AI results into district-level interventions like cool-roof mandates and green-corridor zoning.

B. Integration of Explainable AI (XAI) for Public Accountability

A critical novelty of this study is the conceptual exploration of Explainable AI (XAI) in the context of urban heat. In municipal planning, transparency is as vital as accuracy. This work introduces a framework for using XAI to justify resource allocation, ensuring that AI-driven cooling strategies are defensible, transparent, and resistant to algorithmic bias, particularly when addressing the needs of vulnerable socio-economic demographics.

C. Socio-Ecological Framing

Most UHI reviews treat "sustainability" as a synonym for human thermal comfort. This research extends the conceptual boundaries of UHI mapping to include ecological resilience. The work offers a dual-benefit framework that protects both urban residents and local wildlife from thermal stress, a perspective currently underrepresented in machine learning literature by linking ML detection to biodiversity preservation and habitat connectivity.

2 Significance of AI in UHI mapping and analysis

Urban Heat Islands (UHIs) exacerbate energy costs, ecological disruption, and heat-related morbidity [42, 117]. Traditional mapping often fails to capture the non-linear complexities of urban microclimates, however, but Artificial Intelligence (AI) acts as a transformative catalyst by processing high-dimensional datasets including satellite

imagery, meteorological records, and land-cover data with unprecedented speed and spatial precision [74].

Urban Heat island (UHI) is one of the major problems in cities, and it increases the costs of energy, disrupts natural harmony, and worsens heat-related illnesses [42, 43],Wong, Paddon, and Jimenez 2013). AI is exhibiting as a game-changing tool for UHI mapping and making tasks easier and more accurate to overcome these obstacles. The potential of Artificial Intelligence (AI) to quickly and accurately interpret huge volumes of diverse data, including satellite images, weather reports, and land cover information, is crucial to UHI mapping [74]. AI technologies like deep learning and machine learning models examine these datasets to trace complex spatial patterns and relationships to allow the creation of UHI maps for detailed spatial resolution. Three primary capabilities of AIwith UHI Mapping and Analysis include:

- **Pattern Recognition & Forecasting:** Machine learning (ML) and deep learning architectures identify intricate spatial relationships and temporal dynamics, enabling real-time monitoring and predictive modelling of UHI severity [79].
- **Evidence-Based Governance:** AI provides stakeholders with the empirical basis needed for targeted adaptive management. This facilitates the protection of vulnerable populations and the mitigation of environmental injustices [129] by generating high-resolution thermal maps.
- **Interdisciplinary Synergy:** AI-driven frameworks foster collaboration across data science, urban planning, and remote sensing, bridging the gap between raw data and sustainable mitigation strategies [47].

AI shifts UHI analysis from descriptive mapping to proactive sustainability analysis, ensuring cities become more resilient and liveable through precise, data-driven interventions. UHI mapping with AI makes it easy to track and forecast real-time urban temperature differences [79]. Timely interventions and adaptive management methods are implemented to address heat-related concerns by frequently assessing the incoming data streams with AI algorithms to detect changes in UHI severity and spatial distribution [45, 114]. Policymakers, urban planners, and community stakeholders obtain evidence-based decision-making with AI-generated UHI maps to offer significant insights. These findings offer better efforts to safeguard at-risk populations, level the playing field in terms of environmental injustice, and advance urban equality and social justice [12, 126]. Data science, urban planning, meteorology, and remote sensing benefit from AI-driven UHI mapping for multidisciplinary research projects and collaborations among academic institutions, government organisations, businesses, and civil society. The significance of artificial intelligence (AI) for mapping urban heat islands (UHIs) lies in its ability to enhance our understanding of UHI dynamics, support evidence-based decision-making, and facilitate the implementation of effective mitigation strategies [9, 47]. Cities are made more sustainable, resilient, and livable for present and future generations by utilising AI to better understand UHI dynamics, increase urban planning techniques, and more. Key aspects of AI in UHI mapping include:

a) *Accuracy and Precision:* AI algorithms analyse large volumes of heterogeneous data, including satellite imagery, meteorological data, and land cover information, with high accuracy and precision. The extraction with AI models provides complex spatial

patterns and relationships to generate finer spatial resolution UHI maps with greater detail than traditional methods.

b) *Efficiency and Scalability:* AI-driven UHI mapping reorganises the process of data processing, analysis, and model development for improved efficiency and scalability. AI techniques like machine learning and deep learning provide rapid analysis of vast amounts of data for timely UHI mapping across large geographic areas and at different spatial and temporal scales.

c) *Real-time Monitoring and Prediction:* AI-powered UHI mapping facilitates real-time monitoring and prediction of temperature variations within urban areas. The continuous analysis of incoming data streams with AI approaches detects changes in UHI intensity and spatial distribution to offer timely interventions and adaptive management strategies to address heat-related risks.

d) *Equity and Social Justice:* AI-driven UHI mapping contributes to equity and social justice by identifying areas with high heat exposure and vulnerability within cities. The impact of UHI highlights disparities among different groups, so AI-generated UHI maps offer targeted interventions to protect vulnerable communities and reduce environmental inequalities.

e) *Innovation and Collaboration:* The application of AI in UHI mapping fosters innovation and collaboration across diverse disciplines, including remote sensing, climatology, urban planning, and data science. Interdisciplinary research initiatives and partnerships between academia, government agencies, industry, and civil society are essential for advancing AI-driven UHI mapping and addressing complex urban heat challenges.

The importance of AI in urban heat island mapping stems from its capacity to improve our comprehension, tracking, and reduction of UHI effects on cityscapes, public health, and community welfare. The AI capabilities can be used to build more resilient, sustainable, and liveable cities can be built for present and future generations.

2.1 Effects of artificial intelligence in mapping urban heat Islands and mitigations

, Artificial Intelligence (AI) has a significant impact on mapping urban heat islands (UHIs) and implementing mitigations [11, 101]. The following are cases of artificial intelligence implementation and the resulting impacts:

a) *Accurate Mapping:* AI algorithms like machine learning and deep learning models have enabled more accurate and efficient mapping of UHIs. These algorithms process a substantial volume of satellite imagery, aerial photographs, and other spatial data to identify and delineate UHI zones with higher precision than traditional methods.

b) *Predictive Modelling:* AI-based predictive models forecast future UHI patterns based on various factors such as land use changes, urban development, and climate projections. This helps urban planners and policymakers anticipate potential hotspots and prioritise mitigation efforts.

c) *Identification of Hotspots:* AI algorithms analyse diverse data sources to identify specific hotspots within cities. This information assists in targeting mitigation strategies to areas with the utmost need.

d) *Mitigation Strategy Optimisation:* AI techniques optimise the design and implementation of mitigation strategies intended to reduce UHI effects. For example,

AI algorithms analyse the effectiveness of green infrastructure like parks and green roofs in mitigating UHIs by considering factors like vegetation type, placement, and size.

e) *Energy Efficiency:* AI-powered tools optimise energy consumption in urban areas by providing insights into how buildings and infrastructure contribute to UHIs. The analysis of energy usage patterns and building characteristics can recommend strategies for improving energy efficiency and reducing heat emissions.
f) *Urban Planning:* AI facilitates data-driven urban planning with the integration of UHI mapping for decision-making processes. Planners use AI-generated insights to inform zoning regulations, transportation planning, and the placement of new developments to minimise UHI effects and enhance overall urban livability.
g) *Public Health Impact:* AI-enabled UHI mapping helps to identify areas with higher health risks due to extreme heat for assisting targeted interventions such as the provision of cooling centres and public health outreach programs to vulnerable populations.
h) *Community Engagement:* AI tools enhance community engagement by visualising UHI data in accessible formats to request feedback from residents. This participatory approach empowers communities to contribute to UHI mitigation efforts and fosters collaboration between stakeholders.

The application of artificial intelligence is of critical importance in the process of precisely mapping urban heat islands (UHIs), projecting future trends, locating hotspots, optimising mitigation techniques, improving energy efficiency, influencing decisions on urban design, addressing public health concerns, and promoting community engagement [16, 27]. These developments lead to the advancement of strategies for more effective urban heat island management in urban contexts.

2.2 Artificial intelligence for sustainable development

Artificial Intelligence (AI) act as an innovative operating tool for sustainable development across various sectors to offer advanced solutions for global challenges [96]. The vast amounts of data and advanced algorithms are used for more informed decision-making processes that prioritise environmental conservation, social equity, and economic prosperity [8, 99, 131]. AI facilitates more accurate monitoring and management of natural resources in the sphere of environmental sustainability for water and forests with techniques like remote sensing and predictive analytics [41]. AI enables systems and tools to optimise energy efficiency for reduced emissions to mitigate climate change impacts with optimised power grids to enhance renewable energy integration in promoting smart solutions [81, 95]. In terms of social sustainability, AI promotes inclusivity and equality by providing access to education, healthcare, and economic opportunities for marginalised communities through personalised learning platforms, telemedicine, and job-matching algorithms. AI contributes to economic sustainability by driving innovation, improving productivity, and fostering entrepreneurship through automation, data-driven insights, and predictive modelling [2, 107]. The potential of AI for sustainable development can be harnessed to address ethical considerations, such as algorithmic bias, data privacy, and equitable access to technology, to ensure that AI initiatives promote justice, transparency, and human rights. AI acts as a tool for sustainable

development, and societies can strive towards achieving the United Nations Sustainable Development Goals to create a more resilient, equitable, and thriving future for all.

2.3 Artificial intelligence for ecological conservation

The use of artificial intelligence (AI) is becoming increasingly recognised as an essential instrument for ecological conservation. AI provides novel approaches to tackle difficult environmental challenges and safeguard biodiversity [54, 97]. Conservationists are provided with real-time data and insights that can be used to inform conservation strategies through technologies that are driven by artificial intelligence (AI). These technologies enable more efficient and precise monitoring of ecosystems, wildlife populations, and habitat changes [56, 119]. Machine learning algorithms evaluate big datasets from sources including satellite photography, camera traps, and audio recordings to identify species, detect unlawful actions such as poaching and deforestation, and track shifts in ecological patterns over time [113, 132]. Artificial intelligence makes it possible to conduct predictive modelling to forecast the effects of climate change, human encroachment, and invasive species on fragile ecosystems. This makes it possible for conservation efforts to be proactive and adaptable [4, 108]. Drones and autonomous vehicles are augmented with artificial intelligence to offer new capabilities for conducting surveys and patrolling protected areas [22, 93]. Advanced analytics optimise conservation interventions such as the restoration of habitats and the planning of wildlife corridors. The integration of artificial intelligence (AI) into ecological conservation initiatives improves the efficiency, scalability, and sustainability of conservation efforts, ultimately protecting the priceless natural heritage of our planet for future generations and the stakeholders [87].

2.4 Heat Island mitigations, sustainable development and ecological conservation

The mitigation of heat islands is a critical component of both sustainable development and ecological conservation. Sustainable development aims to balance economic growth, social equity, and environmental protection to meet current needs while preserving the ability of future generations to meet their own needs. Mitigation strategies for heat islands align closely with these objectives by reducing energy consumption, improving air quality, enhancing urban livability, and protecting ecosystems.

Sustainable urban planning practices with increased green spaces, implementation of cool roofs and pavements, and promotion of energy-efficient building designs support the mitigation of heat islands by reducing surface temperatures and cooling the surrounding environment [51, 110]. These strategies alleviate the negative impacts of extreme heat on human health and well-being but also support ecological conservation efforts with the preservation of biodiversity, enhancing habitat connectivity, and restoring natural ecosystems within urban landscapes. Green infrastructure, such as parks, urban forests, and green roofs, also helps mitigate heat islands by acting as natural heat sinks, absorbing heat, sequestering carbon, and providing wildlife habitats [25, 56, 82]. The integration of green infrastructure into urban design and development plans provides mitigation of heat islands in promoting ecological conservation and enhancing urban biodiversity. The prevention of heat islands is of the utmost importance for sustainable development and ecological conservation because it tackles the interrelated concerns of urbanisation, climate change, and the loss of biodiversity [19, 54, 123]. Cities can offset the negative effects of heat islands by implementing a variety of measures

that prioritise environmental sustainability, social fairness, and economic resilience. Sustainable transportation initiatives like increasing public transit, promoting walking and cycling, and reducing vehicular emissions make significant contributions to heat island mitigation with the decrease in the heat generated from transportation infrastructure and promoting low-carbon modes of transportation [53]. These initiatives not only mitigate the urban heat island effect but also decrease air pollution, ease traffic congestion, and enhance the overall quality of life for residents [30, 32]. This will allow cities to cultivate communities that are healthy and resilient while also protecting the natural ecosystems that are necessary for life to exist anywhere on Earth.

3 Policy support and promotion strategies for heat Island mitigations

A supportive policy environment for promoting stakeholder engagement can accelerate the adoption of heat island mitigations to create more resilient, sustainable, and equitable urban environments [64, 121]. Effective heat island mitigations require policy support and promotion strategies. Governments at the local, regional, and national levels have a vital role to play in creating frameworks that incentivise and mandate the adoption of such measures [18, 64]. Building codes and regulations that require cool roofing materials, reflective pavements, and energy-efficient building designs are one key policy approach. Governments also offer financial incentives like tax credits or grants to encourage property owners to invest in heat island mitigation technologies [10, 116, 127]. Public-private partnerships facilitate collaboration between government agencies, businesses, non-profit organisations, and community groups to develop and implement comprehensive mitigation strategies. Public awareness campaigns and educational initiatives raise awareness about the significance of heat island mitigation and empower individuals and communities to take action.

3.1 Strategies for mapping urban heat Islands

Mapping urban heat islands (UHIs) requires a multifaceted approach that integrates various data sources, spatial analysis techniques, and advanced technologies. Satellite remote sensing with thermal sensors is a primary method for capturing thermal imagery and identifying temperature variations across urban areas [68, 124]. Geographic Information Systems (GIS) are then employed to process and analyse the thermal data alongside other geospatial information, such as land cover, land use, and socioeconomic factors [102, 103]. Machine learning algorithms, including neural networks and random forest models, are increasingly utilised to automate the detection and classification of UHIs from satellite imagery, enabling rapid and accurate mapping at large scales [57]. Ground-based measurements, such as weather stations and mobile sensors, provide validation and calibration for remote sensing data, enhancing the accuracy and reliability of UHI mapping efforts. Citizen science initiatives and participatory mapping approaches engage local communities in data collection and interpretation for a collaborative approach towards UHI mapping and empowering residents to contribute to mitigation efforts [28]. These strategies can be used by researchers and urban planners to generate comprehensive UHI maps for informed, evidence-based decision-making to prioritise mitigation interventions and promote the creation of healthier and more resilient cities.

3.2 Policy support for mapping urban heat Islands

Policy support for mapping urban heat islands (UHIs) is essential for evidence-based decision-making and prioritising mitigation efforts in urban areas. Governments play a crucial role in facilitating UHI mapping by establishing regulations to mandate the collection and dissemination of thermal data, requiring building developers to conduct heat island assessments as part of environmental impact assessments [23, 35]. Policymakers can allocate funding and resources to support research initiatives aimed at advancing UHI mapping techniques and technologies, including the development of high-resolution satellite imagery and the deployment of ground-based sensor networks [79]. Governments can provide a collaboration between public agencies, academic institutions, and private sector stakeholders to share data, expertise, and best practices for UHI mapping and mitigation [37, 130]. The integration of UHI mapping with urban planning processes and decision-making frameworks enables policymakers to prioritise investments in green infrastructure, sustainable building design, and heat mitigation strategies that enhance the resilience, livability, and sustainability of cities in the face of climate change and urbanisation pressures.

3.3 Promotion strategies for mapping urban heat Islands

Promotion strategies for mapping urban heat islands (UHIs) are crucial to raising awareness, adopting collaboration, and encouraging widespread adoption of UHI mapping techniques among stakeholders [38, 83, 86]. Public awareness campaigns play a vital role in educating communities about the impacts of UHIs on public health, energy consumption, and urban resilience, highlighting the importance of accurate mapping for effective mitigation strategies. Engaging with local governments, urban planners, and policymakers is essential to integrate UHI mapping into urban planning processes and decision-making frameworks [17, 86]. Workshops, seminars, and training programs provide stakeholders with the knowledge and skills required to collect, analyse, and interpret thermal data with advanced technologies and spatial analysis techniques [1, 3]. Partnerships between academia, industry, and civil society organisations facilitate knowledge exchange, data sharing, and collaborative research initiatives aimed at advancing UHI mapping methodologies and addressing data gaps in underserved communities [100, 128]. Leveraging digital platforms and open-access datasets enhances accessibility and transparency, empowering communities to participate in UHI mapping efforts and contribute local knowledge and observations [16, 49, 85]. Through the promotion of a multi-stakeholder approach for UHI mapping and mitigation, stakeholders collectively identify the priority areas to implement targeted interventions and monitor progress towards creating healthier, more resilient, and sustainable urban environments.

3.4 Sustainable development and ecological conservation with urban heat Islands mapping

UHI mapping is a powerful tool for promoting sustainable development agendas, supporting ecological conservation initiatives, and creating healthier, more resilient, and equitable urban environments for current and future generations [18, 64]. Cities maintain and restore natural habitats, improve biodiversity, and promote ecosystem resilience by incorporating ecological conservation principles into urban planning and design processes with the use of UHI mapping data [24, 98]. UHI mapping helps in the

identification of deprived populations excessively impacted by excessive heat, and later it offers a path for the adoption of measures focused on equity to protect public health and enhance social resilience. Sustainable development and ecological conservation efforts benefit significantly from the mapping of urban heat islands (UHIs) [19, 118]. Sustainable development goals, such as ensuring access to clean energy, promoting sustainable cities and communities, and combating climate change, are directly supported by UHI mapping initiatives [40, 98]. The development of green infrastructure interventions like planting trees, creating urban green spaces, and incorporating cool roofing materials allows cities to reduce surface temperatures, lower energy consumption for cooling, improve air quality, and enhance overall livability with ecological conservation.

3.5 Artificial intelligence for sustainable development and ecological conservation

Artificial Intelligence (AI) stands as a powerful partner for sustainable development and ecological conservation. The processing of vast amounts of data to derive actionable insights offers decision-making across various sectors to promote environmental stewardship, social equity, and economic prosperity [99, 106]. AI offers a pivotal role in sustainable development for optimising resource management, enhancing energy efficiency, and fostering innovation in renewable technologies [80]. AI-driven predictive analytics estimates environmental trends, enabling proactive measures to mitigate climate change impacts and manage natural resources more effectively [36, 92, 104]. AI-powered smart systems enable precise monitoring and control of energy consumption, water usage, and waste generation in urban environments to contribute to the creation of sustainable cities and communities. AI empowers researchers and conservationists to monitor and protect biodiversity more efficiently for ecological conservation [19, 39]. Machine learning algorithms analyse ecological data from remote sensing, camera traps, and acoustic recordings to identify patterns, track wildlife populations, and detect habitat disturbances [13, 75]. AI helps in predictive modelling to assess the vulnerability of species to environmental threats, informing conservation strategies and prioritising interventions. AI for sustainable development and ecological conservation can be used to achieve the United Nations Sustainable Development Goals for safeguarding the natural heritage for future generations.

4 Challenges and opportunities for mapping urban heat Islands with Artificial Intelligence

Mapping urban heat islands (UHIs) with artificial intelligence (AI) presents both challenges and opportunities in understanding and addressing urban heat challenges [34, 84]. Challenges include data availability and quality, model interpretability, and equity considerations. Obtaining comprehensive and reliable datasets for training AI models remains a hurdle, while ensuring the fairness and transparency of AI-driven UHI mapping is essential for addressing environmental inequalities [55, 125]. Urban heat islands (UHIs) present significant challenges to urban sustainability, but mapping them with artificial intelligence (AI) offers both barriers and avenues for progress.

Challenges:

a) *Data Complexity:* Accessing comprehensive and standardised datasets, including satellite imagery and meteorological data, provides a challenge due to data heterogeneity and quality discrepancies.

b) *Interpretability:* AI models used for UHI mapping, often complex and opaque, lack interpretability, hindering understanding of the underlying factors driving UHI formation.
c) *Equity Concerns:* Biases in AI algorithms intensify disparities, and vulnerable communities may be disproportionately affected by this.
d) *Generalisation:* AI models trained on specific datasets may struggle to generalise to diverse urban environments, impacting the scalability and applicability of UHI mapping efforts.

Opportunities:

a) *Enhanced Precision:* AI enables precise analysis of diverse data sources to offer a more nuanced understanding of UHI dynamics at various scales.
b) *Real-time Monitoring:* AI facilitates real-time monitoring of UHI patterns to enable timely interventions to mitigate heat-related risks.
c) *Innovation:* AI fosters innovation in interdisciplinary research to promote collaboration between fields such as climatology, urban planning, and data science.
d) *Community Engagement:* AI-driven UHI mapping empowers communities through participatory approaches, promoting equity and resilience.

AI-driven UHI mapping has the potential to inform evidence-based decision-making, encouraging sustainable urban development, and minimising the impacts of UHIs on public health and well-being.

5 Recommendations and future research prospects

Enhancing data accessibility and quality, integrating multiple data sources, and developing explainable AI methodologies are crucial steps in improving the accuracy and resilience of urban heat island (UHI) mapping with AI [20, 73]. Integrating AI models into decision-making frameworks and validating them through benchmarking can contribute to the advancement of UHI mapping [40, 63, 98]. The success of AI-driven UHI mapping depends on long-term monitoring, community involvement, and capacity-building efforts to create more sustainable and livable urban environments [33, 105, 130]. Recommendations and prospects for future research affecting the mapping of urban heat islands with artificial intelligence are:

a) *Enhanced Data Accessibility and Quality:* Future research should focus on improving the accessibility and quality of data relevant to UHI mapping, including satellite imagery, meteorological data, and ground truth measurements. Efforts to develop open-access data repositories and standardised data collection protocols facilitate collaboration and advance AI-driven UHI mapping efforts.
b) *Integration of Multi-Modal Data Sources:* Research should explore methods for integrating multi-modal data sources, such as remote sensing imagery, urban infrastructure data, and socioeconomic indicators, to enhance the accuracy and robustness of AI models for UHI mapping. Techniques for fusing heterogeneous data streams and incorporating temporal dynamics provide valuable insights into UHI dynamics over time.
c) *Development of Explainable AI Models:* Given the complexity of AI models used for UHI mapping, future research should prioritise the development of explainable AI

techniques to enhance the interpretability and transparency of model predictions. Explainable AI methods help stakeholders understand the factors driving UHI patterns and build trust in AI-driven decision-making processes.

d) *Scale-Up and Transferability of AI Models:* Efforts to scale up AI models for UHI mapping to cover larger geographic areas and diverse urban environments are essential for addressing global urban heat challenges. Research should focus on developing transferable AI models that can be generalised across different regions and climatic conditions to facilitate broader adoption and application.

e) *Validation and Benchmarking:* Future research emphasises the validation and benchmarking of AI models for UHI mapping with standardised evaluation metrics and ground truth datasets. Comparative studies assessing the performance of different AI algorithms under various scenarios provide valuable insights into the strengths and limitations of different modelling approaches.

f) *Integration with Decision Support Systems:* AI-driven UHI mapping can be integrated into decision support systems and urban planning frameworks to inform evidence-based decision-making and prioritise UHI mitigation strategies. Collaborative efforts between researchers, policymakers, and urban planners are essential for translating UHI mapping results into actionable policies and interventions.

g) *Long-Term Monitoring and Adaptation:* Research should focus on establishing long-term monitoring programs to track changes in UHI intensity and spatial distribution over time. Continuous monitoring using remote sensing technologies and ground-based sensors can provide valuable data for assessing the effectiveness of UHI mitigation measures and adapting management strategies to evolving urban and climatic conditions.

h) *Community Engagement and Capacity Building:* Engaging local communities and building capacity among stakeholders are critical for the success of AI-driven UHI mapping initiatives. Future research should prioritise community engagement efforts, including participatory mapping exercises, citizen science initiatives, and outreach programs, to raise awareness about UHI risks and empower communities to participate in UHI mitigation efforts.

More accurate, practical, and fair solutions to reduce heat-related hazards and encourage sustainable urban development can be achieved if these suggestions and opportunities for future study are adequately considered in the area of artificial intelligence mapping of urban heat islands.

6 Conclusion

This article has provided a comprehensive conceptual framework for the integration of Machine Learning (ML) into Urban Heat Island (UHI) detection and sustainability analysis. The study argues that AI is not merely a technical tool, but a systemic catalyst for transitioning from reactive urban management to proactive, climate-resilient governance with the synthesis of diverse technological and planning discourses. The primary contribution of this work is the development of an interdisciplinary workflow that bridges the gap between raw geospatial data and inclusive urban policy. This has demonstrated that theoretically how ML architectures such as Convolutional Neural Networks (CNNs) for spatial pattern recognition and Ensemble Learning for feature importance can be operationalised to identify the complex, non-linear drivers of thermal stress. This

conceptual framework aligns UHI mapping as a foundation for SDG 11 (Sustainable Cities and Communities) to shift the focus from simple temperature measurement to the analysis of socio-ecological vulnerability. This conceptual model promotes a "Precision Mitigation" approach from a planning perspective and policymakers can move beyond generalised cooling strategies by adopting the AI-driven roadmap like :

- **Thematic Integration:** Incorporating thermal dynamics into the design of ecological corridors and biodiversity conservation plans.
- **Environmental Justice:** Utilising ML to highlight the intersection of thermal hotspots and socio-economic deprivation, ensuring that cooling interventions (e.g., green roofs, urban canopies) are prioritised in under-resourced landscapes.
- **Infrastructural Optimisation:** Providing a theoretical basis for "Cool-City" policies that emphasise energy-efficient building designs and high-albedo materials in precisely identified high-risk zones.

7 Future directions

This study provides the foundation for several future research trajectories as a conceptual work. There is a clear need for comparative case studies to validate this framework across different global climatic zones, ranging from arid urban centers to tropical megacities. Furthermore, future research should explore the integration of Digital Twin technology and IoT-sensor networks into this framework to create real-time, 4D urban climate models. The adoption of these AI-driven approaches represents a promising frontier for safeguarding ecological resilience and urban livability in an era of rapid urbanisation and climatic instability.

Author contributions
Dr. Nick P. Bassill (NB), and Dr. Deepak Kumar (DK) conceived and designed the framework of the research, and Dr. Deepak Kumar (DK), performed the research. Dr. Nick P. Bassill (NB), along with Dr Deepak Kumar (DK) analyzed the data with contributions to the editorial input. Conceptualization, methodology and formal analysis: NB, DK; investigation: NB; visualization: NB, DK; writing—original draft: DK; writing—review and editing: DK; All authors read and approved the final manuscript.

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The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Declaration of Generative AI/AI-assisted technologies in the writing process
During the preparation of this work, the first author used Microsoft Copilot for creative support and ChatGPT for grammar and spelling review to improve the English language. Following the use of these tools, the authors carefully reviewed and edited the content as necessary and assumed full responsibility for the final content of the publication.

Ethics approval and consent to participate
Not applicable.

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